

TBA4245 GEODESY

Version 29.1.2018

Example exercise no. 1:**GPS baselines. Baseline transformation and accuracy measures****Assignment text****Some hints to the tasks:**

- Skogseth/Holsen: Compendium in Basic GPS
- See also chapters 3 and 10 in the GNSS book (Hofmann-Wellenhof).
- The transformation programs SkTrans or GISLINE Trans or similar (including Holsen programs) can be used in the calculations.
- The manuals of the programs Gemini and GIS/Line (but in Norwegian...)

Task no. 1. Transformations of satellite baselines

The denominations are generally in meter and gon, if no other denominations are specified.

As a check of the network of control points in Trondheim municipality, some GPS baselines were measured (only GPS satellites used).

The baselines Ladehammeren–Holtermannsveien (L-H), Solemsvåttan–Holtermannsveien (S-H) and Solemsvåttan–Ladehammeren (S-L) were measured by use of GPS and calculated by using a post processing program. Geocentric Cartesian coordinates, datum WGS84:

From	To	ΔX	ΔY	ΔZ
L	H	3541,713	-811,048	-1666,943
S	H	1084,518	-9212,999	-87,373
S	L	-2457,183	-8401,958	1579,590

The Norwegian mapping authority (Kartverket) states the following coordinates and heights of the control points, given in the horizontal datum EUREF89-UTM-Zone 32 and the vertical datum NN2000:

Point	N	E	H_{ell}	H_{NN2000}	$N = H_{ell} - H_{NN2000}$
S	7032158,662	579058,469	462,071	422,853	39,218
L	7036209,878	571150,913	108,875	69,634	39,241
H	7032450,406	569796,193	110,462	71,006	39,456

Geoidal height model

In the actual area the deflection of the plumb line (the vertical) is calculated to be constant: +3,02 mgon in the northern direction and +1.73 mgon in the eastern direction. The geoidal height (geoid-ellipsoid separation) is found to be $N = 39,218$ m in the control point Solemsvåttan (in NN2000, horizontal datum EUREF89).

Use the GPS measurements to carry out the following calculations:

- (a) Calculate the local geodetic coordinates of the GPS baselines from Solemsvåttan to Holtermannsveien and from Ladehammeren to Holtermannsveien. (Transformation to a local terrestrial horizontal coordinate system.)
Which easy/simple calculations can be done to check if the transformation is OK? Carry out this check.

- (b) Calculate the distance and the bearing from Solemsvåttan to Holtermannsveien in the map projection plane. Compare with values calculated directly from the EUREF89 UTM coordinates.
- (c) **Either:** Calculate the zenith angle from Solemsvåttan to Holtermannsveien from given heights and coordinates. Compare with the zenith angle calculated from the GPS measurements.
Or, probably easier: Calculate ΔH (orthometric height difference) from the GPS measurements and compare to the ΔH in the table above.

Task no. 2. Accuracy measures of the measured GPS baselines

The GPS baseline from S to H will be used. See task no. 1 for coordinates etc.

The processing of the baselines gives the correlation matrix $\mathbf{K}$ of the baseline S-H and the standard deviations on ΔX , ΔY , ΔZ in the Cartesian geocentric coordinate system:

$$\mathbf{K} = \begin{pmatrix} 1,0000 & 0,1307 & 0,5335 \\ 0,1307 & 1,0000 & 0,1441 \\ 0,5335 & 0,1441 & 1,0000 \end{pmatrix}$$

$$s_{\Delta X} = 0,5 \text{ mm} \quad s_{\Delta Y} = 0,4 \text{ mm} \quad s_{\Delta Z} = 1,0 \text{ mm} \quad \leftarrow \text{Unrealistic low values?}$$

- (a) The 3D Cartesian X , Y , Z coordinate system
 Calculate the values of the variance/covariance matrix of ΔX , ΔY , ΔZ of the baseline S-H.
- (b) From the 3D Cartesian X , Y , Z to the local geodetic coordinate system, x , y , z
 Calculate the values of the variance/covariance matrix of x , y , z for the baseline S-H.
- (c) From the local geodetic coordinate system, x , y , z to the decomposed observations in the UTM map projection plane and height difference or zenith angle referred to the geoid (the datum NN2000)
 State with equations how the variance/covariance matrix of the decomposed observations of a GPS baseline (distance, bearing, height difference or zenith angle) can be calculated.
 Calculate the values of the variance/covariance matrix of the decomposed observations for the baseline S-H.
- (d) Often empirical values are used for the standard deviations.
 Use assumed empirical values of the GPS observations and calculate the variances of the distance, bearing and the zenith angle of both of the baselines.
 State with equations how the complete variance/covariance matrix can be calculated.
 Evaluate the methods and the results of the 2 calculations in (a) and (b).
 How will you calculate the optimal accuracy measures of the measured baselines?

Task no. 3. Checking the results in 1 and 2 with the program GISLINE

- (a) The calculations above can be tested by using the baseline transformation program in for instance the Norwegian land surveying programs GIS/Line or Gemini GPS/NET.
 See also the compulsory assignment no. 1 covering this item.)